Research Statement

I am an applied economist working in industrial organization and the economics of innovation. My research asks how and why firms allocate their research: the mode of that research, why they enter certain areas, and how government assistance and changes in market structure affect the scale and composition of their portfolios. I combine applied theory with applied econometrics, using production and innovation surveys, patents, award records, and merger files.

Internal and External R&D: An analysis of costs and benefits. I study in-house and contracted-out R&D among Dutch ICT firms from 2000 to 2020. Estimating a dynamic discrete-choice model with mode-specific investment costs, I find that contracted-out R&D costs about 2.45 million euros, versus 0.13 million for in-house R&D, which is why so few firms contract out alone. Doing both is cheaper than the sum of the two costs: in-house R&D offsets almost all of the extra cost of going outside. Productivity gains are similar across modes. A uniform subsidy raises R&D participation and firm value by more than a subsidy aimed only at in-house R&D.

Entry into Innovation Areas. I then ask where firms put that research. The paper studies multi-area entry by listed U.S. electronics firms. I apply Latent Dirichlet Allocation to over 630,000 patents to define twenty innovation areas and merge them with market-implied valuations, then estimate a static entry game with moment inequalities. Firms enter areas where expected patent valuations are higher. Holding those valuations fixed, additional rivals reduce expected profit. Firms gain from occupying more areas at once. A large yearly per-area cost and rival congestion still limit expansion. From observed portfolios, a 25 percent reduction in that cost raises to 22–27 percent the share of firms that occupy an extra area.

The Impact of Government Financial Assistance on the Scope and Direction of Firm Innovation, with Costanza Cincotta. Governments spend hundreds of billions of dollars each year financing private innovation. We ask whether this spending expands, redirects, or reallocates research. We combine a Salop model with a stacked difference-in-differences on U.S. public firms from 2000 to 2021, matching Good Jobs First awards to patent data, and compare recipients with close and distant rivals defined by text-based product-market overlap. Distant rivals provide the preferred reading of the treated firm's own response. Recipients' patenting scale rises by about 3 percent relative to distant product-market controls. Close product-market rivals do not change how much they patent. Direction shifts toward the most-cited technologies, and the shift is strongest for cash grants.

Mergers and Innovation: An Exploration of Scope and Direction of Innovation, with Sabien Dobbelaere and José Luis Moraga-González. How do mergers change the scale and composition of firm innovation? We study U.S. public–public mergers from 1980 to 2020 using combined acquirer–target patent portfolios. A compact framework isolates business stealing, knowledge pooling, and activation, and keeps product-market overlap distinct from technological overlap. Comparing treated merger pairs to matched control pairs, joint-portfolio scope is larger after mergers, by about 0.18 on the annual measure. The difference is larger when pre-merger technologies overlap, especially among non-horizontal deals. Joint portfolios also show higher alignment with highly cited patents.

These papers treat the allocation of research as a set of discrete choices over modes, areas, and portfolios. Costs, rivals, cash, and ownership then move both how much firms invent and what they invent.

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